Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A device that converts energy of combustion of fuels like hydrogen and methane, directly into electrical energy is known as:

(A) Electrolytic cell

(B) Dynamo

(C) Ni-Cd cell

(D) Fuel cell

Q.2

Which of the following statement is correct?

(A) Cathode is -ve terminal in both, glavanic and electrolytic cells.

(B) Anode is +ve terminal in both, galvanic and electrolytic cells.

(C) Cathode and anode are -ve terminal in electrolytic and galvanic cell.

(D) Cathode and anode are +ve terminal in electrolytic and galvanic cell.

Q.3

A concentration cell is a galvanic cell in which

(A) decrease in free energy in a spontaneous chemical process appears as electrical energy

(B) decrease in free energy in a spontaneous physical process appears as electrical energy

(C) decrease in free energy in a spontaneous physical or chemical process appears as electrical energy

(D) a non-spontaneous physical or chemical process produces electrical energy.

Q.4

Among the following cells:
(I) Leclanche cell
(II) Nickel-Cadmium cell
(III) Lead storage battery
(IV) Mercury cell primary cells are

(A) I and II

(B) I and III

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(C) II and III

(D) I and IV

Q.5

How long a current of 3 amperes has to be passed through a solution of $AgNO_3$ to coat a metal surface of $80\ cm^2$ and $0.005\ mm$ thick layer. Density of Ag is $10.5\ g\ cm^{-3}$

(A) 125.1 seconds

(B) 12.5 seconds

(C) 155.2 seconds

(D) 200 seconds

Q.6

In a hydrogen-oxygen fuel cell, combustion of hydrogen occurs to

(A) produce high purity water

(B) create potential difference between two electrodes

(C) generate heat

(D) remove adsorbed oxygen from electron surfaces

Q.7

In a fuel cell methanol is used as fuel and oxygen gas is used as an oxidizer. The reaction is
$CH_3OH(l) + 3/2O_2(\ g) \longrightarrow CO_2(\ g) + 2H_2O(l)$
At $298\ K$ standard Gibb's energies of formation for $CH_3OH(l), H_2O(l)$ and and $CO_2(g)$ are $-166.2 - 237.2$ and
$-394.4\ kJ\ mol^{-1}$ respectively. If standard enthalpy of combustion of methonal is $-726\ kJ\ mol^{-1}$, find the efficiency (in %) of the fuel cell :

(A) 97

(B) 95

(C) 99

(D) 93

Q.8

Based on the cell notation for a spontaneous reaction, at the anode :
$Ag(s)|AgCl(s)|Cl^-(aq)||Br^-(aq)\ |Br_2(l)|\ C(s)$

(A) AgCl gets reduced

(B) Ag gets oxidized

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(C) Br^- gets oxidized

(D) Br_2 gets reduced

Q.9

Which colourless gas evolves, when NH_4Cl reacts with zinc in a dry cell battery

(A) NH_4

(B) N_2

(C) H_2

(D) Cl_2

Q.10

The e.m.f. of a Daniell cell at $298\ K$ is E_1. Zn $\left|\begin{matrix}ZnSO_4\\(0.01M)\end{matrix}\right|\left|\left|\begin{matrix}CuSO_4\\(1.0M)\end{matrix}\right|\right.Cu$

When the concentration of $ZnSO_4$ is $1.0M$ and that of $CuSO_4$ is $0.01M$, the e.m.f. changed to E_2. What is the relationship between E_1 and E_2 ?

(A) $E_2 = 0 \neq E_1$

(B) $E_1 > E_2$

(C) $E_1 < E_2$

(D) $E_1 = E_2$

Q.11

The standard potentials of $Ag^+/Ag, Hg_2{}^{2+}/2Hg, Cu^{2+}/Cu$ and Mg^{2+}/Mg electrodes are $0.80, 0.79, 0.34$ and $-2.37\ V$, respectively. An aqueous solution which contains one mole per litre of the salts of each of the four metals is electrolyzed. With increasing voltage, the correct sequence of deposition of the metals at the cathode is

(A) Ag, Hg, Cu, Mg

(B) Cu, Hg, Ag only

(C) Ag, Hg, Cu only

(D) Mg, Cu, Hg, Ag

Q.12

$1\ 0\ L$ each of a buffer containing $1\ mole NH_3$ and $1\ mol$ of NH_4^+ were placed in the cathodic and anodic half-cells and $965C$ of electricity was passed. If anodic and cathodic half cells reactions involve oxidation and reduction of water only as

$2H_2O \longrightarrow 4H^+ + O_2 - 4e^-$

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$2H_2O + 2e^- \longrightarrow H_2 + 2OH^-$

Then pH of

(A) cathodic solution will increase

(B) anodic solution will decrease

(C) both the solutions will remain practically constant

(D) both the solutions will increase

Q.13

1.08 g of pure silver was converted into silver nitrate and its solution was taken in a beaker. It was electrolysed using platinum cathode and silver anode. 0.01 Faraday of electricity was passed using 0.15 volt above the decomposition potential of silver. The silver content of the beaker after the above shall be

(A) 0 g

(B) 0.108 g

(C) 1.08 g

(D) None of these

Q.14

The Zn acts as sacrificial or cathodic protection to prevent rusting of iron because :

(A) E°_{OP} of $Zn < E^\circ_{OP}$ of Fe

(B) E°_{OP} of $Zn > E^\circ_{OP}$ of Fe

(C) E°_{OP} of $Zn = E^\circ_{OP}$ of Fe

(D) Zn is cheaper than iron

Q.15

"Maintenance free" batteries now in use in place of common batteries have

(A) Electrode made of lead-lead oxide

(B) Electrodes made of calcium-containing lead alloy

(C) Non aqueous solvents as medium

(D) Platinum electrodes

Q.16

Aqueous solution of which of the following compounds is the best conductor of electric current?

(A) Acetic acid, $C_2H_4O_2$

(B) Hydrochloric acid, HCl

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(C) Ammonia, NH_3

(D) Fructose, $C_6H_{12}O_6$

Q.17

A hydrogen gas electrode is made by dipping platinum wire in a solution of HCl of $pH = 10$ and by passing hydrogen gas around the platinum wire at one atm pressure. The oxidation potential of electrode would be ?

(A) $0.59\ V$

(B) $0.118\ V$

(C) $1.18\ V$

(D) $0.059\ V$

Q.18

An electrochemical cell is set up as:
$Pt; H_2(1\ atm)\ |HCl(0.1M)||CH_3COOH(0.1M)|\ H_2$ (1atm); Pt.
The e.m.fof this cell will not be zero, because

(A) the temperature is constant

(B) e.m.f depends on molarities of acids used

(C) acids used in two compartments are different

(D) pH of $0.1MHCl$ and $0.1MCH_3COOH$ is not same

Q.19

Mark the false statement?

(A) A salt bridge is used to eliminate liquid junction potential

(B) The Gibbs free energy change, ΔG is related with electromotive force E as $\Delta G = -nFE$

(C) Nernst equation for single electrode potential is $E = E^o - \frac{RT}{nF} \log a_{M^{n+}}$

(D) None of these

Q.20

In the silver plating of copper, $K\ [Ag(CN)_2]$ is used instead of $AgNO_3$. The reason is

(A) a thin layer of Ag is formed on Cu

(B) more voltage is required

(C) Ag^+ ions are completely removed from solution

(D) less availability of Ag^+ ions, as Cu cannot displace Ag from $[Ag(CN)_2]^-$ ion

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Q.1 Answer:

Fuel cell

Solution:

A device that converts energy of combustion of fuels, directly into electrical energy is known as fuel cell.

Q.2 Answer:

Cathode and anode are -ve terminal in electrolytic and galvanic cell.

Solution:

Cathode and anode are $(-)$ ve terminals in electrolytic and galvanic cell.

Q.3 Answer:

decrease in free energy in a spontaneous physical process appears as electrical energy

Solution:

In concentration cell the spontaneous process is physical in nature involving transfer of matter from higher concentration to lower concentration in indirect manner

Q.4 Answer:

I and IV

Solution:

Primary cells are those cells, in which the reaction occurs only once and after use over a period of time, it becomes dead and cannot be reused again. e.g., Leclanche cell and mercury cell.

Q.5 Answer:

125.1 seconds

Solution:

Volume = Area $\times$ thickness Mass = Volume $\times$ density $\therefore$ Mass of Ag to be deposited
$= \frac{80 \times 0.005}{10} \times 10.5 = 0.42$ g
Amount deposited $= \frac{i \times t \times E.wt}{96500}$
$\therefore 0.42 = \frac{108 \times 3 \times \mathrm{t}}{98500}$ $\therefore \mathrm{t} = 125.1$ **seconds**

Q.6 Answer:

create potential difference between two electrodes

Solution:

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In $H_2 - O_2$ fuel cell, the combustion of H_2 occurs to create potential difference between the two electrodes

Q.7 Answer:

97

Solution:

$$CH_3OH(l) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$
$$\Delta G_r = [\Delta G_f(CO_2, g) + 2\Delta G_f(H_2O, \ell)] -$$
$$\left[\Delta G_f(CH_3OH, \ell) + \frac{3}{2}\Delta G_f(O_2, g)\right]$$
$$= -394.4 + 2(-237.2) - (-166.2) - 0$$
$$= -394.4 - 474.4 + 166.2 = -702.6\ \text{kJ}$$
$$\%\ \textbf{efficiency} = \frac{702.6}{726} \times 100 = 97\%$$

Q.8 Answer:

Ag gets oxidized

Solution:

Ag becomes oxidized and Br^- becomes reduced.

Q.9 Answer:

H_2

Solution:

$$2NH_4Cl + Zn \rightarrow 2NH_3 + ZnCl_2 + H_2 \uparrow.$$

Q.10 Answer:

$E_1 > E_2$

Solution:

Cell reaction is, $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF}\ell n\frac{[Zn^{2+}]}{[Cu^{2+}]}$$

Greater the factor $\left[\frac{(Zn^{2+})}{(Cu^{2+})}\right]$**, less is the EMF Hence** $E_1 > E_2$

Q.11 Answer:

Ag, Hg, Cu, Mg

Solution:

More the reduction potential, more is the deposition of metals at cathode. Cation having E^0 value less than -0.83 V (reduction potential of H_2O) will not deposit from aqueous solution. Hence

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correct order of A deposition of the metal at the cathode is $\mathrm{Ag} > \mathrm{Hg} > \mathrm{Cu} > \mathrm{Mg}$

Q.12 Answer:

both the solutions will remain practically constant

Solution:

Due to buffer action the pH **will remain practically constant.**

Q.13 Answer:

$0\ \mathrm{g}$

Solution:

$\mathrm{Ag}^{+} + \mathrm{e}^{-} \longrightarrow \mathrm{Ag}$

$1\ \mathrm{F} = 1$ **mole of electrons** $= 96500\mathrm{C}$

$0.01\ \mathrm{F} = 1.08\ \mathrm{gAg};\ \mathrm{Ag}$ **left** $= 1.08 - 1.08 = 0\ \mathrm{g}$

Q.14 Answer:

$\mathrm{E}^{\circ}_{\mathrm{OP}}$ of $\mathrm{Zn} > \mathrm{E}^{\circ}_{\mathrm{OP}}$ of Fe

Solution:

$\mathrm{E}^{\circ}_{\mathrm{OP}}$ **of** $\mathrm{Zn} > \mathrm{E}^{\circ}_{\mathrm{OP}}$ **of** Fe

Q.15 Answer:

Electrodes made of calcium-containing lead alloy

Solution:

Calcium-tin-lead alloys are used for making grids for maintenance-free batteries. The calcium content in such alloys for positive grids has varied generally from about 0.06 **to about** 0.1% **by weight of the alloy while the tin has generally ranged from about** 0.1 **up to** 0.8% **and even more. More typically, the calcium content in such alloys when used for making maintenance-free battery grids has been at least about** 0.08% **by weight or more.**

Q.16 Answer:

Hydrochloric acid, HCl

Solution:

HCl **completely dissociates to give** H^{+} **and** Cl^{-} **ions, hence act as very good electrolyte. While others are non- electrolytes.**

Q.17 Answer:

$0.59\ \mathrm{V}$

Solution:

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$H_2 \longrightarrow 2H^+ + 2e^-$

1 atm 10^{-10}

$E_{H_2/H^+} = 0 - \frac{0.059}{2} \log \frac{\left(10^{-10}\right)^2}{1}$

$E_{H_2/H^+} = +0.59\ V$

Q.18 Answer:

pH of $0.1MHCl$ and $0.1MCH_3COOH$ is not same

Solution:

For a concentration cell having different concentrations of ions. $E = -\frac{0.0591}{n} \log \frac{c_1}{c_2}$

If all the concentrations are identical then obviously the cell voltage is zero. But as the pH **of** $0.1MHCl$ **(strong acid) and** pH **of** $0.1MCH_3COOH$ **is (weak acid) not same, therefore the cell voltage is not zero.**

Q.19 Answer:

Nernst equation for single electrode potential is $E = E^o - \frac{RT}{nF} \log a_{M^{n+}}$

Solution:

Correct Nernst equation is

$E = E^o + \frac{2.303RT}{nF} \log a_{M^{n+}}$

Q.20 Answer:

less availability of Ag^+ ions, as Cu cannot displace Ag from $[Ag(CN)_2]^-$ ion

Solution:

In the silver plating of copper, $K[Ag(CN)_2]$ **is used instead of** $AgNO_3$**. Copper being more electropositive readily precipitate silver from their salt solution**

$Cu + 2AgNO_3 \longrightarrow Cu(NO_3)_2 + Ag$

whereas in $K[Ag(CN)_2]$ **solution a complex anion** $[Ag(CN)_2]^-$ **is formed and hence** Ag **are less available in the solution and therefore copper cannot displace Ag from its complex ion.**

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